

Demo: Locating Leakage Current via Magnetometers

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Abstract

Electrical leakage is pervasive: ground faults account for roughly 80% of distribution-network faults, and in homes, leakage routinely trips residual-current devices. But these devices report only that leakage exists somewhere downstream, leaving the actual fault location to manual, circuit-by-circuit testing. We present a non-contact method that sweeps a magnetometer across a wall, tracks it with visual-inertial odometry, and jointly estimates cable positions along with their leakage currents. The key is separating the dominant load current from the far weaker leakage current, which we do by exploiting their distinct spatial signatures. Our prototype detects 91% of injected-leakage cases.

CCS Concepts

• **Computer systems organization** → **Sensors and actuators**; • **Hardware** → *Sensor applications and deployments*; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Keywords

magnetometer sensing, electrical safety, leakage current, residual-current device, visual-inertial odometry

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1 Introduction

Leakage current escapes its intended circuit, often through degraded insulation, and can energize exposed conductive surfaces, creating a shock hazard [7, 14]. It is a common symptom of electrical faults. In distribution networks, single-phase ground faults, in which current leaks to earth, account

for roughly 80% of all faults [8]. In buildings, electrical regulations and standards commonly require protection by residual-current device (RCD)¹ [13, 17]. This device detects the leakage and disconnects the circuit. Such RCD trips are among the most frequent household electrical problems [4, 6].

However, an RCD trips only when leakage creates a line-neutral current imbalance above a threshold, and it indicates only that the leak is somewhere downstream. Finding the leak still requires opening the panel, isolating circuits one by one with a clamp meter, and performing insulation tests [9, 15, 18]. This lack of location information has practical consequences: of the first 13 US states to adopt the 2020 National Electrical Code (NEC), 11 weakened or removed its new GFCI requirement in response to nuisance trips [5].

Our goal is non-contact localization of leakage current, enabling electricians and maintenance personnel to find faults without touching potentially hazardous equipment. Commodity non-contact voltage testers use capacitive coupling to find live wires behind walls but provide no information about current leakage [20]. ElectriSense and MagnifiSense non-intrusively identify device activity from electromagnetic signals, rather than localizing wiring or leakage faults [11, 21]. Recent work localizes leakage from magnetic line-neutral imbalance, but requires modifying the panel by installing a transformer and injecting a tone at the neutral point [3]. These approaches do not meet our goal.

Contribution. We bridge this gap using a low-cost, off-the-shelf magnetometer to sense magnetic fields from live conductors. At AC frequency, the quasi-static magnetic field passes through nonmagnetic wall materials with little perturbation. Healthy and leaking cables exhibit distinct spatial signatures in both radial decay and angular rate. At practical detection distances, however, the load-current field dominates the leakage-current field. We therefore use a moving aperture to separate them and locate the leaking cable.

Demo Video: <https://youtu.be/Qb3FGjQ0x0E>

2 Sensing Method

Line (L) and neutral (N), separated by d along $\hat{x}$, carry opposing current magnitudes I_L along $+\hat{z}$ and I_N along $-\hat{z}$ (Fig. 1a). We call their nearly equal magnitude $I \approx I_L \approx I_N$ the load

¹Ground-fault circuit interrupters (GFCIs) in US terminology.



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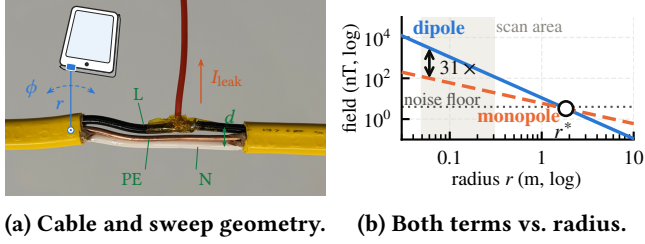


Figure 1: The red lead injects the leak; the 5.5-mm line-neutral dipole outweighs it 31× at 6 cm.

current and their difference $I_{leak} = I_L - I_N$ the leakage current. In a healthy cable, $I_L = I_N$, so their net current is zero; their individual magnetic fields nevertheless combine into the dipole term. If leakage returns *outside the cable* rather than through its protective-earth (PE), a fault leaves $I_{leak} \neq 0$; this residual current produces a monopole term. At radius $r \gg d$ and azimuth ϕ , the total magnetic field $\vec{B}$ is [10]:

$$\vec{B} = \vec{B}_{dip} + \vec{B}_{mono} = \frac{\mu_0 I d}{2\pi r^2} \begin{pmatrix} -\sin 2\phi \\ \cos 2\phi \end{pmatrix} + \frac{\mu_0 I_{leak}}{2\pi r} \begin{pmatrix} -\sin \phi \\ \cos \phi \end{pmatrix}. \quad (1)$$

The dipole and monopole terms differ in radial decay ($1/r^2$ vs. $1/r$) and angular rate (2ϕ vs. ϕ).

Since $|I_{leak}| \ll I$, the monopole is a small perturbation near the cable: for $d = 5.5$ mm, $I = 10$ A, and $I_{leak} = 30$ mA, at 6 cm the dipole is 31× larger (Fig. 1b). One might move farther for the monopole to dominate. However, their crossover gives $r^* = Id/I_{leak} = 1.8$ m, where the leak is only 3.3 nT, below the noise floor most sensors reach in a whole second [16].

Instead, we operate closer to the cable, where both fields are stronger but mixed. The user sweeps the sensor over radius and angle while visual-inertial odometry (VIO) supplies poses. We fit Eq. (1) by least squares and solve for r , Id , and I_{leak} . We declare a cable as leaking if we find $I_{leak} > 15$ mA, below the tripping point for RCDs.

The fit assumes a stable load during each sweep and no nearby energized cable. To estimate leakage position, the user scans multiple points along the cable: adjacent healthy and leaking scans bracket the fault between them, narrowing but not pinpointing its location.

3 Implementation

Our testbed uses a 25 ft 12/2 Type NM-B cable [2]. At its midpoint, we tap one conductor through a decade resistance box to introduce a controlled 0–50 mA leak while the cable carries a 2–10 A, 60 Hz load current. For magnetic sensing, we mount a MEMSIC MMC5983MA module [16] on an 11-inch iPad Pro (11th generation) running ARKit for VIO [1]. The magnetometer samples continuously at 1 kHz with 800 Hz bandwidth and automatic SET/RESET disabled. A Raspberry Pi 4B [19] receives the measurements

from the magnetometer over UART, applies lock-in detection to extract the 60 Hz component, and streams the result to the iPad over USB Ethernet [12]. For each sweep, we move the sensor at a 5–50 cm radial distance around the suspected cable location. Experiments assume a stable load and no nearby live cable. Our open-source implementation of the offline least-squares field estimator is at <https://github.com/MachW-at-UW-Madison/wirecheck>.

4 Results

Across all 17 scans (11 leaking and 6 healthy), the prototype detects 10 of the 11 leaking scans with a 90.9% true-positive rate, missing only the 10 mA trial. Six healthy scans remain below the threshold, giving 16/17 (94.1%) overall accuracy.

Overall fit quality. We sample the magnetic field at known 24 positions around a cable carrying a 10 A load with a 30 mA injected leak. We plot the estimated fields due to load and leak currents in Fig. 2a (assuming known d to decouple Id). At each sample, we define the normalized magnitude residual as $(|\vec{B}_{meas}| - |\vec{B}_{model}|)/|\vec{B}_{meas}|$. Across the aperture, the median residual is 1.06% and the worst-case residual is 12.2%.

Recovery across leak levels. At a fixed 5 A load, we sweep injected leakage from 0 to 50 mA in 10 mA steps to evaluate recovery in Fig. 2b. Below roughly 10 mA, leakage is not reliably separable from sensor noise, pose uncertainty, and current drift. We therefore set a 15 mA threshold to tell healthy from faulty cables, below the typical 30 mA RCD trip level while retaining sensitivity to safety-relevant leaks.

Real-time iPad sweeps. We sweep the iPad freehand around a cable carrying a 5 A load, once healthy and once with a 40 mA injected leak, and refit as each quarter-second capture arrives (Fig. 2c). Before about 80 captures, the estimates are unstable and fluctuate by tens of milliamps. After roughly 180 captures (45 s), the faulted estimate converges to 40 mA and the healthy estimate falls below 15 mA. We therefore withhold the live verdict until the estimates stabilize.

Locating the cable. Across the same 17 scans (2.5–10.0 A, 0–50 mA, 15–72 captures), the fitted cable centres lie within 1.7 mm of their per-setup means (Fig. 2d). This measures cable-localization repeatability, not fault-location accuracy; Sec. 2’s bracketing approach only narrows the fault to the interval between adjacent scans.

5 Demonstration

The demo requires at least a clear 5 m × 2 m area away from the venue’s wiring. *Safety.* The test cable is not connected to mains. A signal generator and fused, low-voltage power amplifier drive the 60 Hz load current into a resistive load bank. At the cable midpoint, a resistance box diverts 0–50 mA through a dedicated return wire to the source, so neither earth nor exposed metal forms the return path. All energized

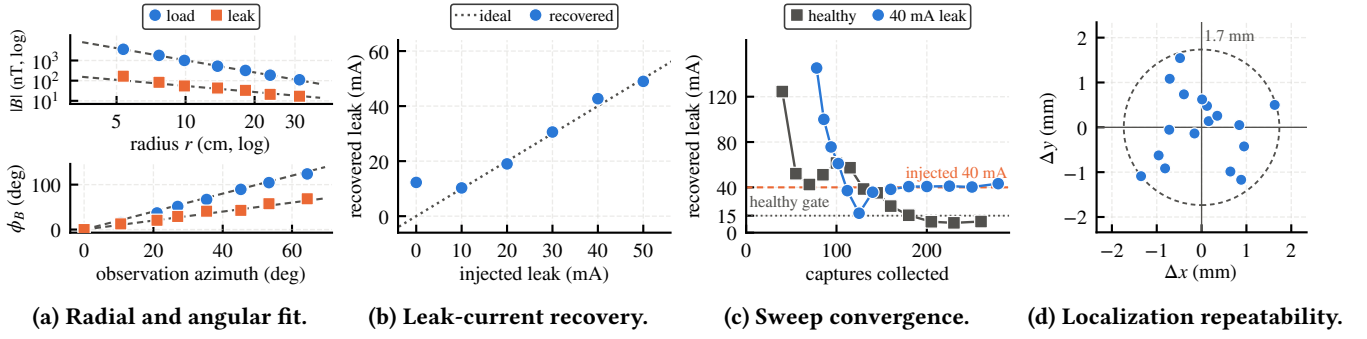


Figure 2: Experimental evaluation. (a) A 24-position scan matches the radial and angular model under a 10 A load and 30 mA leak. (b) Recovery tracks 0–50 mA injected leakage at 5 A. (c) Healthy and 40 mA leakage that iPad sweeps converge after 180 captures. (d) 17 scans localize the cable within 1.7 mm.

components remain enclosed, and only demo presenters adjust the leakage current. *What visitors see.* Visitors sweep the iPad across the cable while a display updates the fitted cable cross-section (not the fault position along the cable), load current, and leakage estimate. The verdict remains provisional until the estimates stabilize after roughly 40–60 s.

6 Conclusion

We presented a low-cost, non-contact system that locates a cable, detects and estimates leakage from VIO-tracked magnetic sweeps, despite a load current dominating the leakage signal. These results point toward practical leakage localization without contact with hazardous equipment. Future work will relax the stable-load and isolated-cable assumptions, and test main cables with real loads and leakages behind walls.

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